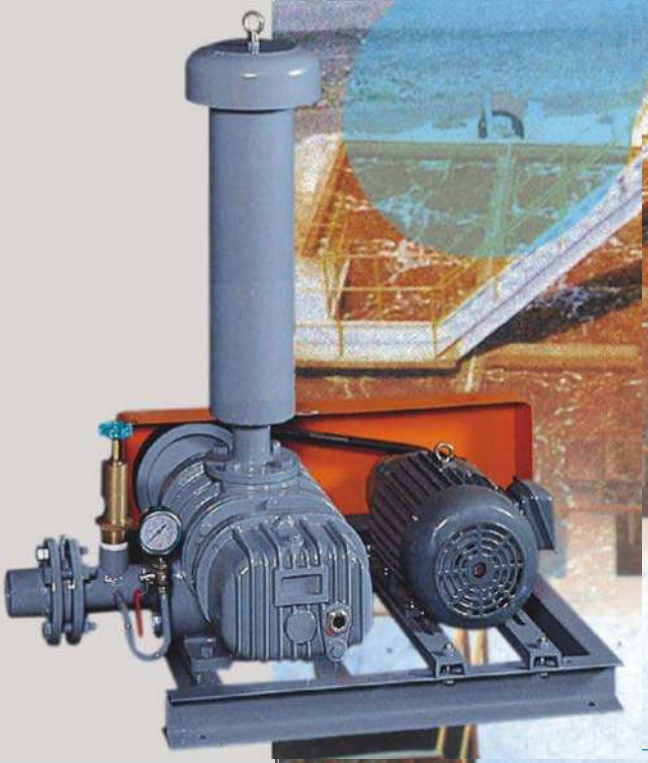
perotac
MADE IN TAIWAN

Roots Blowers
Membrane Disc Diffuser

ISO 9001
BSMI

CE

ISO-9001/CNS12681





THREE LOBES ROOTS BLOWER

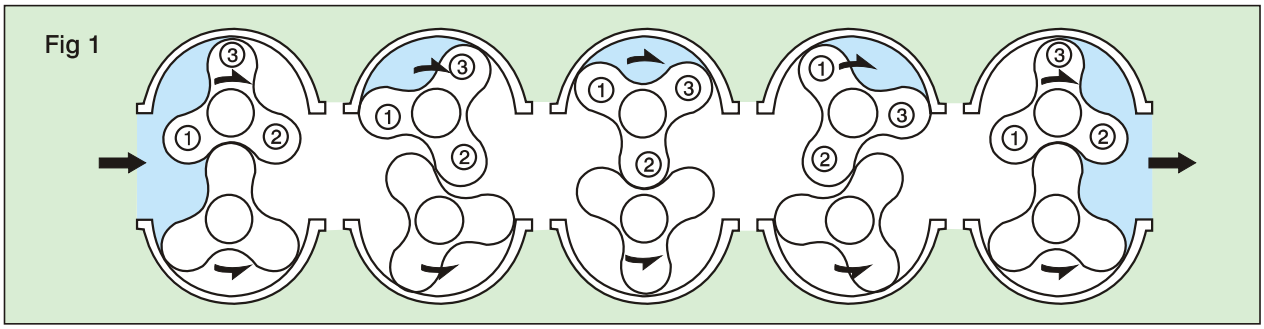
GENERAL

SHOWFOU 3 LOBES ROOTS BLOWERS, Know also as POSITIVE DISPLACEMENT ROTARY LOBE PUMPS, are used in conditions ranging from strong vacuum to high pressure in all branches of industry, especially in waste water treatment plant. The simple design, easy handling, and stable performance make possible a wide range of applications. Their main characteristic is to produce a constant flow of gas at different pressure and efficiencies are normally high.
Another important feature is that the pumped gas does not get contaminated because there are no lubricants in the area in which the compression occurs.
The blowers consist of a cast-iron casing in which two rotors with conjugate shape rotate and are sync-hronized by a set of timing gears. During the rotation, there is no contact either between the lobes or between the lobes and the casing.
Two front and rear side covers close each casing : the bearing of the shafts and the sealing devices to avoid gas leakage are located there.

OPERATING PRINCIPLE

The two three-lobe rotors rotate one opposite to the other; their movement is synchronized by a set of timing gears.
The geometric configuration of the lobes allows them to have, at any moment, a generatrix in common; due to very low machine tolerance, this common generatrix acts also as a seal device for both rotors. During the operation, one or two lobes of each rotor come into contact with the internal surface of the casing : this create a chamber in which the gas is trapped.
As the rotation continues, the trapped gas moves along until it reaches the position of discharging port. The two rotors make six intake and exhaust cycles per revolution; therefore the capacity of the SHOWFOU roots blower is determined by its operating speed and increase proportionally with the speed, but is independent from the pressure differential between inlet and outlet ports.

Fig 1

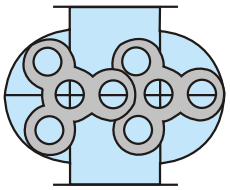


ADVANTAGES OF 3 LOBES DESIGN

Owing to new 3 lobes' rotor design, the reverse flow pressure variation period is only 2/3 of that of a conventional 2 lobes' rotor, the peak pressure value is also lower. So the noise and pressure pulsation are greatly limited.
Advantages of SHOWFOU 3 LOBES ROOTS BLOWER are as follows :

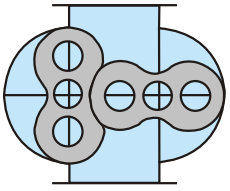
1. Less back flow, steady air flow rate, lower vibration and noise.
2. Discharge pressure pulsation are reduced, so lighter loading of bearings and timing gears ensures the long servicing life.
3. Running on the same operating speed, the 3 lobes blower can deliver large air flow and volume than conventional 2 lobes type.
4. Adequate clearance between the rotors and the rotors with the casing ensure no contact during operation, and this make the efficiency even higher.
5. The precision of rotors is fully controlled and variation of precision between blowers is almost nil because the rotors are produced by utilizing a precision NC machine.
6. The rotors are dynamically balanced in the fabrication stage already, so these rotors are almost free from vibrations ◦

3-Lobes Cylinder



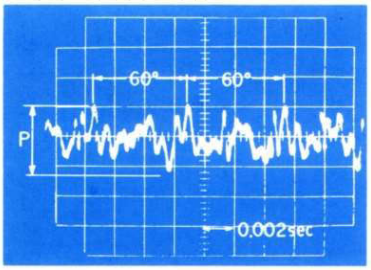
(Fig.1)

2-Lobes Cylinder

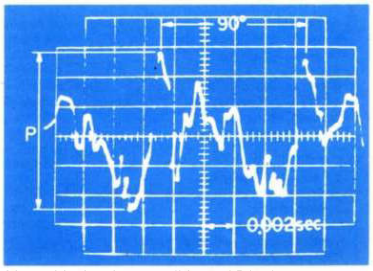


(Fig.2)

Discharge Pressure Variation in a 3-Lobes Rotor



Discharge Pressure Variation in a 2-Lobes Rotor



Note: Under the condition of Discharge Pressure at 5000 mmAq and Revolution of 1560 r.p.m. for both blowers.

CHARACTERISTICS

RL



RW



RECOMMENDED PRESSURE OPERATION RANGE :
RL: 1000~8000mmAq RW: 8000~10000mmAq

■ MAXIMUM PERFORMANCE

1 atm = 10333mmAq = 760 mmHg

	SINGLE STAGE		DOUBLE STAGE	
TYPE	RL	RW	RL	RW
MAXI. PRESSURE	10,000mmAq (1.0kg/cm^2)	12,000mmAq (1.2kg/cm^2)	15,000mmAq (1.5kg/cm^2)	20,000mmAq (2.0kg/cm^2)
MAXI. VACUUM	-6,700mmAq (-500mmHg)	-8,100mmAq (-600mmHg)	-8,100mmAq (-600mmHg)	-8,800mmAq (-650mmHg)

■ STRUCTURE ANALYSIS

	GREASE LUBRICATION	OIL LUBRICATION	WATER JACKET FOR COOLING
FRONT BEARING	RW	RL	RW
REAR BEARING	–	RL、RW	RW
GEAR BOX	–	RL、RW	RW

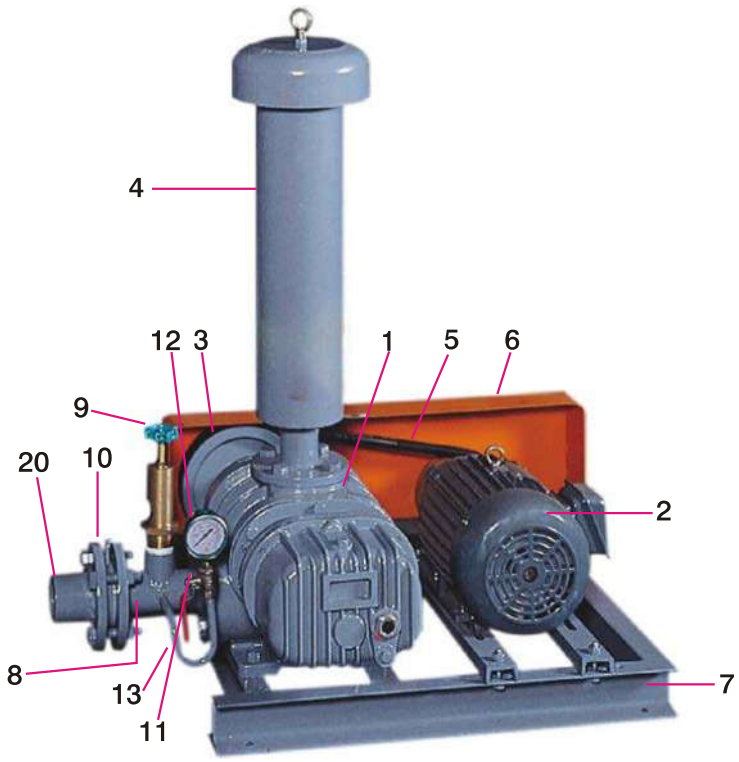
SPECIFICATIONS

RL / RW : Blower Body ONLY.
RLC/RWC : Complete Set. - Blower Body With Standard Accessories For Pressure Operation.
RLV/ RWV : Complete Set. - Blower Body With Standard Accessories For Pressure Operation.
Please refer to page 3 for the picture of complete set.

TYPE	BORE (mm)	BLOWER BODY	COMPLETE SET FOR PRESSURE OPERATION	COMPLETE SET FOR VACUUM OPERATION
RL	40	RL-40	RLC-40	–
	50	RL-50	RLC-50	RLV-50
	65	RL-65	RLC-65	RLV-65
	80	RL-80	RLC-80	RLV-80
	100	RL-100	RLC-100	RLV-100
	125	RL-125	RLC-125	RLV-125
	150	RL-150	RLC-150	RLV-150
	200	RL-200	RLC-200	RLV-200
	250	RL-250	RLC-250	RLV-250
	300	RL-300	RLC-300	RLV-300
RW	50	RW-50	RWC-50	RWC-50
	65	RW-65	RWC-65	RWV-65
	80	RW-80	RWC-80	RWV-80
	100	RW-100	RWC-100	RWV-100
	125	RW-125	RWC-125	RWV-125
	150	RW-150	RWC-150	RWV-150
	200	RW-200	RWC-200	RWV-200
	250	RW-250	RWC-250	RWV-250
	300	RW-300	RWC-300	RWV-300

COMPLETE SET ROOTS BLOWER INSTALLATION : with standard accessories

■ PRESSURE OPERATION (RLC 、RWC)



STANDARD ACCESSORIES

NO	NOAME
1	BLOWER BODY
2	MOTOR
3	BLOWER PULLEY
4	SUCTION SILENCER SET (WITH AIRFILTER & COVER)
5	V-BELT
6	BELT COVER
7	BASE PLATE
8	T-TYPE REDUCER
9	PRESSURE SAFETY VALVE
10	CHECK VALVE
11	BALL VALVE
12	PRESSURE GAUGE
13	GAUGE PIPE
20	DISCHARGE FLANGE (OPTIONAL)

RL PERFORMANCE TABLE (FOR PRESSURE OPERATION)

■ RLC - Complete Set ■ RL - Blower Body only

TYPE	Bore	Revolution	1000mmAq		2000mmAq		3000mmAq		4000mmAq		5000mmAq		6000mmAq		7000mmAq		8000mmAq	
			9.8kpa		19.6kpa		29.4kpa		39.4kpa		49.0kpa		58.8kpa		68kpa		78.4kpa	
	Inch(mm)	R.P.M.	m³/min	kw	m³/min	kw	m³/min	kw	m³/min	kw	m³/min	kw	m³/min	kw	m³/min	kw	m³/min	kw
40	1½(40)	800	0.70	0.25	0.56	0.50	0.45	0.77	0.35	0.94	0.26	1.14	0.21	1.34	0.15	1.52	0.10	1.70
		900	0.85	0.28	0.71	0.56	0.61	0.81	0.50	1.05	0.42	1.27	0.36	1.50	0.30	1.71	0.26	1.91
		1000	1.01	0.31	0.87	0.62	0.75	0.90	0.67	1.16	0.58	1.43	0.50	1.67	0.46	1.90	0.40	2.12
		1100	1.17	0.35	1.03	0.68	0.92	0.99	0.82	1.34	0.73	1.57	0.67	1.84	0.61	2.09	0.56	2.34
		1200	1.32	0.36	1.19	0.74	1.08	1.08	0.98	1.40	0.88	1.71	0.83	2.01	0.77	2.29	0.72	2.45
		1300	1.48	0.41	1.34	0.80	1.23	1.17	1.14	1.52	1.05	1.86	0.98	2.18	0.93	2.47	0.88	2.56
		1400	1.64	0.44	1.51	0.86	1.40	1.26	1.29	1.63	1.22	1.99	1.14	2.34	1.08	2.66	1.04	2.66
		1500	1.80	0.47	1.66	0.92	1.56	1.35	1.45	1.75	1.36	2.14	1.30	2.51	1.24	2.85	1.20	2.98
		1600	1.96	0.50	1.82	0.98	1.72	1.44	1.61	1.87	1.52	2.28	1.46	2.67	1.41	3.04	1.36	3.30
		1700	2.13	0.57	1.98	1.05	1.87	1.53	1.77	1.99	1.68	2.42	1.61	2.84	1.55	3.23	1.52	3.67
		1800	2.29	0.57	2.14	1.11	2.03	1.62	1.93	2.11	1.83	2.56	1.77	3.01	1.71	3.42	1.67	3.93
		1900	2.45	0.60	2.30	1.20	2.19	1.71	2.09	2.23	1.99	2.70	1.92	3.18	1.86	3.61	1.82	4.26
50	2(50)	800	1.16	0.42	0.92	0.83	0.75	1.29	0.58	1.56	0.44	1.90	0.34	2.22	0.24	2.53	0.17	2.83
		900	1.42	0.47	1.18	0.93	1.00	1.35	0.84	1.75	0.70	2.12	0.60	2.50	0.50	2.85	0.43	3.19
		1000	1.68	0.52	1.45	1.03	1.26	1.50	1.10	1.94	0.96	2.38	0.85	2.78	0.76	3.17	0.68	3.54
		1100	1.94	0.58	1.72	1.13	1.53	1.65	1.37	2.24	1.21	2.62	1.11	3.06	1.02	3.49	0.94	3.90
		1200	2.21	0.63	1.99	1.23	1.80	1.79	1.63	2.33	1.48	2.85	1.37	3.34	1.29	3.81	1.20	4.08
		1300	2.47	0.69	2.26	1.33	2.06	1.95	1.90	2.53	1.74	3.09	1.64	3.62	1.55	4.11	1.46	4.27
		1400	2.73	0.74	2.51	1.44	2.33	2.10	2.15	2.72	2.01	3.32	1.90	3.89	1.81	4.44	1.74	4.44
		1500	3.00	0.79	2.78	1.54	2.59	2.25	2.42	2.92	2.27	3.56	2.17	4.17	2.07	4.76	2.00	4.96
		1600	3.27	0.84	3.04	1.64	2.86	2.40	2.69	3.12	2.54	3.80	2.43	4.45	2.34	5.08	2.27	5.46
		1700	3.55	0.90	3.30	1.75	3.12	2.55	2.94	3.32	2.80	4.03	2.69	4.73	2.59	5.39	2.53	6.12
		1800	3.81	0.95	3.57	1.85	3.39	2.70	3.21	3.52	3.06	4.26	2.95	5.01	2.85	5.70	2.78	6.56
		1900	4.08	1.00	3.84	1.95	3.65	2.85	3.48	3.72	3.31	4.49	3.21	5.29	3.11	6.01	3.04	7.09
65	2½(65)	800	1.95	0.58	1.78	1.14	1.61	1.66	1.46	2.16	1.33	2.62	1.21	3.07	1.12	3.49	1.03	3.91
		900	2.28	0.79	2.11	1.28	1.94	1.86	1.79	2.42	1.65	2.95	1.54	3.45	1.45	3.93	1.35	4.40
		1000	2.63	1.14	2.46	1.42	2.29	2.06	2.12	2.69	1.99	3.28	1.89	3.83	1.78	4.35	1.69	4.89
		1100	2.97	1.29	2.80	1.57	2.62	2.28	2.47	2.96	2.34	3.61	2.23	4.22	2.12	4.81	2.03	5.38
		1200	3.27	1.44	3.14	1.71	2.96	2.49	2.80	3.23	2.68	3.93	2.56	4.60	2.46	5.24	2.36	5.86
		1300	3.63	1.54	3.48	1.85	3.29	2.70	3.14	3.50	3.01	4.26	2.90	4.98	2.80	5.68	2.70	6.35
		1400	3.99	1.64	3.82	1.99	3.64	2.91	3.50	3.77	3.36	4.59	3.24	5.37	3.15	6.12	3.04	6.84
		1500	4.34	1.74	4.16	2.13	3.98	3.11	3.83	4.04	3.70	4.91	3.58	5.75	3.48	6.55	3.38	7.32
		1600	4.67	1.84	4.51	2.27	4.32	3.32	4.17	4.31	4.03	5.24	3.92	6.13	3.82	6.98	3.72	8.28
		1700	5.02	1.94	4.86	2.42	4.66	3.53	4.52	4.57	4.37	5.57	4.26	6.51	4.17	7.42	4.07	8.30
		1800	5.36	2.04	5.20	2.56	5.00	3.73	4.86	4.85	4.71	5.89	4.60	6.89	4.50	7.85	4.41	8.78
		1900	5.71	2.14	5.53	2.70	5.34	3.94	5.19	5.12	5.05	6.22	4.94	7.27	4.84	8.28	4.75	9.27
80	3(80)	800	2.91	1.40	2.55	2.12	2.23	2.81	1.93	3.51	1.67	4.31	1.42	5.43	1.21	6.55	1.02	7.67
		900	3.51	1.50	3.25	2.32	2.93	3.14	2.63	3.84	2.36	4.64	2.12	5.76	1.89	6.88	1.70	8.00
		1000	4.21	1.60	3.95	2.53	3.63	3.47	3.33	4.25	3.06	5.52	2.82	6.64	2.61	7.06	2.42	8.18
		1100	4.91	1.70	4.65	2.75	4.33	3.80	4.03	4.70	3.76	5.71	3.52	6.83	3.31	7.95	3.12	9.07
		1200	5.40	1.80	5.14	2.91	4.82	4.03	4.52	5.04	4.29	6.05	4.07	7.17	3.84	8.29	3.65	9.41
		1300	6.05	1.90	5.75	3.20	5.43	4.25	5.13	5.60	4.95	6.72	4.69	7.84	4.48	8.96	4.29	10.08
		1400	6.53	2.00	6.27	3.47	5.95	4.59	5.65	5.82	5.38	7.05	5.14	8.28	4.93	9.56	4.74	10.81
		1500	7.00	2.10	6.74	3.50	6.42	5.04	6.12	6.27	5.85	7.50	5.61	8.96	5.40	10.42	5.21	11.88
		1600	7.76	2.20	7.50	3.92	7.18	5.26	6.86	6.66	6.61	8.06	6.37	9.52	6.16	10.98	5.97	12.44
		1700	8.26	2.32	8.00	4.15	7.67	5.53	7.36	7.04	7.08	8.53	6.83	10.04	6.61	11.54	6.42	13.04
		1800	8.83	2.42	8.52	4.37	8.17	5.87	7.86	7.49	7.59	8.98	7.33	10.59	7.12	12.21	6.92	13.82
		1900	9.32	2.52	8.97	4.83	8.64	6.20	8.33	7.82	8.06	9.43	7.81	11.15	7.59	12.63	7.39	14.14
100	4(100)	800	5.08	1.37	4.51	2.65	4.00	3.80	3.54	4.88	3.14	5.68	2.79	6.78	2.48	7.62	2.20	8.43
		900	5.91	1.55	5.33	3.00	4.82	4.31	4.36	5.52	3.96	6.55	3.60	7.67	3.29	8.63	3.01	9.54
		1000	7.08	1.83	6.47	3.52	5.93	5.06	5.44	6.48	5.02	7.79	4.64	9.00	4.31	10.13	4.02	11.18
		1100	7.92	2.01	7.31	3.87	6.77	5.56	6.28	7.12	5.86	8.67	5.48	9.90	5.15	11.14	4.75	12.29
		1200	8.76	2.19	8.15	4.22	7.61	6.07	7.12	7.77	6.70	9.55	6.32	10.80	5.99	12.15	5.70	13.41
		1300	9.60	2.38	8.99	4.57	8.45	6.58	7.96	8.42	7.54	10.43	7.16	11.70	6.83	13.17	6.54	14.53
		1400	10.44	2.56	9.83	4.92	9.29	7.08	8.86	9.07	8.38	11.31	8.00	12.60	7.67	14.18	7.38	15.69
		1500	11.34	2.67	10.75	4.93	10.23	7.59	9.75	9.72	9.25	12.22	8.88	13.50	8.56	15.19	8.28	16.75
		1600	12.17	2.89	11.58	5.45	11.05	8.09	10.58	10.37	10.17	12.73	9.80	14.40	9.48	16.20	9.20	17.87
		1700	13.00	3.12	12.41	5.97	11.88	8.60	11.40	11.01	11.00	13.23	10.63	15.30	10.31	17.21	10.03	18.99
		1800	13.83	3.34	13.42	6.50	13.01	9.10	12.53	11.66	12.12	13.87	11.70	16.20	11.37	18.22	11.06	20.11
		1900	14.65	3.57	14.15	7.03	13.75	9.61	13.32	12.31	12.91	14.25	12.54	17.10	12.22	19.23	11.94	21.23

Note: 1. The values are expressed as required power including drive power for the transmission unit.
2. The rated range for the air volume at specified ± 5%.
3. When the design pressure is 8000 mmAq and above, shall be applied RW type (water-cooling design)

RL

RL PERFORMANCE TABLE (FOR PRESSURE OPERATION)

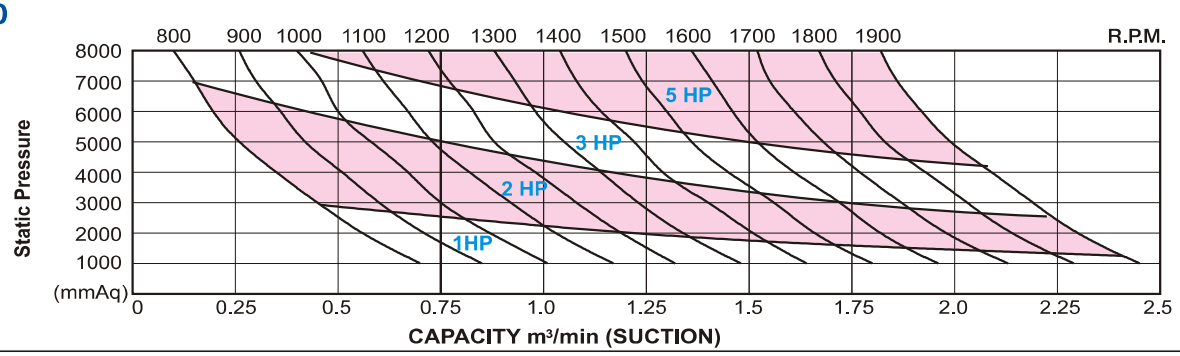
■ RLC - Complete Set ■ RL - Blower Body only

TYPE	Bore	Revolution	1000mmAq		2000mmAq		3000mmAq		4000mmAq		5000mmAq		6000mmAq		7000mmAq		8000mmAq	
			9.8kpa		19.6kpa		29.4kpa		39.4kpa		49.0kpa		58.8kpa		68kpa		78.4kpa	
	inch(mm)	R.P.M.	m³/min	kw	m³/min	kw	m³/min	kw	m³/min	kw	m³/min	kw	m³/min	kw	m³/min	kw	m³/min	kw
125	5(125)	700	9.94	3.31	9.06	5.36	8.38	7.31	7.90	9.35	7.40	11.31	6.91	13.35	6.55	15.31	6.19	17.27
		800	11.52	3.78	10.57	6.07	9.84	8.38	9.32	10.68	8.79	12.98	8.27	15.29	7.86	17.59	7.52	19.89
		900	13.14	4.20	12.29	6.73	11.56	9.36	10.93	11.98	10.40	14.61	9.99	17.13	9.56	19.65	9.19	22.18
		1000	15.05	4.55	14.02	7.45	13.34	10.34	12.74	13.24	12.20	16.03	11.72	18.92	11.79	21.92	10.67	24.82
		1100	16.72	5.10	15.80	8.25	15.10	11.29	14.42	14.55	13.90	17.81	13.38	21.07	12.92	24.98	12.40	28.90
		1200	18.45	5.32	17.51	8.86	16.79	12.31	16.23	15.74	15.74	19.18	15.26	22.93	14.80	26.69	14.30	30.44
		1300	20.11	5.77	19.15	9.62	18.40	13.27	17.87	17.32	17.32	20.75	16.79	24.81	16.37	28.88	15.86	32.19
		1400	21.77	6.23	20.78	10.28	20.01	14.22	19.47	18.27	18.93	22.97	18.38	27.35	17.94	30.63	17.42	33.91
		1500	23.49	6.65	22.50	10.97	21.73	15.30	21.05	19.50	20.51	24.38	20.06	28.82	19.61	35.46	19.08	37.69
		1600	25.27	7.06	24.33	11.59	23.70	16.21	23.06	20.74	22.54	26.33	21.98	30.54	21.59	34.75	21.24	38.97
		1700	27.01	7.38	26.05	12.35	25.32	17.19	24.79	23.21	24.27	27.43	23.73	32.71	23.31	36.92	22.93	41.43
		1800	28.74	8.17	27.79	13.11	27.05	18.18	26.52	24.30	26.00	29.59	25.46	34.87	25.08	39.04	24.71	43.20
150	6(150)	700	13.09	3.99	11.97	6.65	11.25	9.21	10.54	11.77	9.93	14.42	9.41	16.98	8.98	19.54	8.66	22.09
		800	15.06	4.62	13.96	7.59	13.19	10.44	12.41	13.41	11.76	16.38	11.21	19.45	10.71	22.53	10.36	25.60
		900	17.32	4.96	16.22	8.38	15.34	11.81	14.56	15.12	14.01	18.43	13.35	22.06	12.89	25.70	12.35	29.34
		1000	19.54	5.65	18.46	9.33	17.59	13.03	16.93	16.72	16.29	20.41	15.64	23.89	15.09	27.36	14.80	30.95
		1100	21.89	6.15	20.75	10.26	19.84	14.36	19.04	18.47	18.47	22.57	17.78	27.38	17.31	32.20	16.94	37.00
		1200	23.97	6.67	22.88	11.05	21.99	15.53	21.34	19.92	20.69	25.17	20.14	29.55	19.69	33.93	19.33	38.30
		1300	26.16	7.19	24.93	12.58	24.14	16.73	23.35	21.56	22.79	26.95	22.12	31.44	21.67	35.93	21.33	40.42
		1400	28.24	7.69	27.09	12.86	26.18	18.03	25.49	24.07	24.80	28.71	24.22	34.45	23.79	40.19	23.41	46.94
		1500	30.48	8.26	29.32	13.72	28.39	19.31	27.57	25.59	26.99	31.41	26.41	37.22	25.93	43.04	25.57	48.86
		1600	32.83	8.74	31.73	14.59	30.85	20.56	30.07	27.64	29.52	33.16	28.96	37.73	28.52	43.26	28.17	48.79
		1700	34.99	9.19	33.88	15.50	33.00	21.85	32.34	28.79	31.67	35.44	31.11	40.97	30.61	46.41	30.26	52.05
		1800	37.26	9.20	36.16	16.41	35.27	23.29	34.49	29.95	33.94	36.60	33.26	43.26	32.75	49.91	32.39	56.57
200	8(200)	600	23.52	6.77	21.87	11.52	20.52	16.26	19.45	21.29	18.58	25.16	17.61	30.97	16.84	35.81	16.16	40.65
		700	27.90	7.90	26.20	13.40	24.90	18.90	23.80	24.00	22.80	30.00	21.90	37.00	23.00	44.00	22.30	47.00
		800	32.67	9.06	31.04	15.33	29.69	22.17	28.63	27.95	27.66	35.66	26.89	41.45	26.12	47.23	25.45	53.01
		900	37.26	10.06	35.61	17.13	34.26	24.19	33.19	30.97	32.32	38.71	31.45	45.48	30.68	52.26	30.00	59.03
		1000	41.90	11.05	40.29	19.05	38.95	26.67	37.90	35.24	37.05	42.86	36.19	50.48	35.43	58.10	34.76	65.71
		1100	46.70	12.20	45.00	21.00	43.70	30.00	42.60	39.00	41.60	48.00	40.70	56.00	39.90	65.00	39.20	74.00
		1200	50.98	13.25	49.34	23.04	48.10	31.68	47.04	42.24	46.08	50.88	45.22	60.48	44.45	69.12	43.78	79.68
		1300	55.45	14.28	53.77	24.62	52.49	34.47	51.41	45.30	50.42	55.15	49.54	65.00	48.75	75.83	48.06	86.67
		1400	59.93	15.31	58.22	26.19	56.91	38.27	55.80	48.35	54.79	59.42	53.99	70.50	53.08	81.58	52.05	92.66
250	10(250)	600	31.94	8.82	30.00	15.15	28.55	21.78	27.29	29.05	26.13	34.24	25.06	41.50	24.37	46.68	23.71	51.87
		700	37.80	10.29	35.90	17.69	34.30	25.73	33.00	32.16	32.00	40.74	30.90	48.24	29.90	54.67	29.20	61.10
		800	44.24	11.57	42.22	20.14	40.67	28.92	39.52	37.20	38.46	46.50	37.40	53.73	36.43	62.00	35.93	70.27
		900	50.32	12.97	48.39	22.82	46.94	31.12	45.77	41.50	44.52	50.84	43.45	60.17	42.77	69.50	42.10	78.83
		1000	56.48	14.29	54.67	24.50	53.05	34.71	51.90	45.94	50.76	56.19	49.81	66.55	48.95	77.59	48.19	88.63
		1100	63.00	15.75	61.00	27.87	59.40	39.66	58.10	51.46	57.00	62.18	56.00	73.97	55.00	85.76	54.20	97.55
		1200	68.64	17.09	66.82	29.85	65.28	43.22	64.03	56.46	62.88	66.89	61.92	80.28	61.06	92.62	60.19	104.93
		1300	74.65	18.48	72.78	31.67	71.20	46.46	69.92	59.09	68.74	72.85	67.76	86.57	66.87	99.94	66.08	113.36
		1400	80.68	19.86	78.56	34.55	77.05	49.66	75.90	63.71	74.63	78.82	73.63	93.93	72.52	109.08	71.71	124.19
300	12(300)	600	71.2	18.1	68.0	33.8	65.3	48.1	62.6	61.4	60.5	73.6	58.5	84.9	56.8	95.5	55.3	105.2
		700	85.5	20.8	81.4	39.2	78.7	55.9	76.0	71.4	73.9	85.6	71.9	98.8	70.2	111.1	68.7	122.5
		800	98.5	23.7	94.8	44.7	92.1	63.8	89.4	81.4	87.3	97.7	85.3	112.7	83.6	126.8	82.1	139.8
		900	111.5	26.6	108.2	50.1	105.6	71.6	102.9	91.4	100.8	109.7	98.8	126.7	97.1	142.4	95.6	157.1
		1000	124.5	29.6	121.6	57.0	118.9	80.9	116.2	102.9	114.1	123.3	112.1	142.1	110.4	159.6	108.9	176.0
		1100	138.1	34.5	135.1	63.9	132.4	90.2	129.7	114.4	127.6	136.8	125.6	157.6	123.9	176.9	122.4	194.8
		1200	151.7	39.5	148.5	70.8	145.8	99.5	143.1	125.9	141.0	150.4	139.0	171.9	137.3	194.1	135.8	213.7
		1300	165.3	44.5	162.1	83.6	159.4	108.8	156.7	137.4	154.6	164.0	152.6	188.4	150.9	211.3	149.4	232.6
		1400	178.0	47.9	174.6	90.0	171.7	117.2	168.8	148.0	166.5	176.6	164.3	202.9	162.5	227.6	160.9	250.5

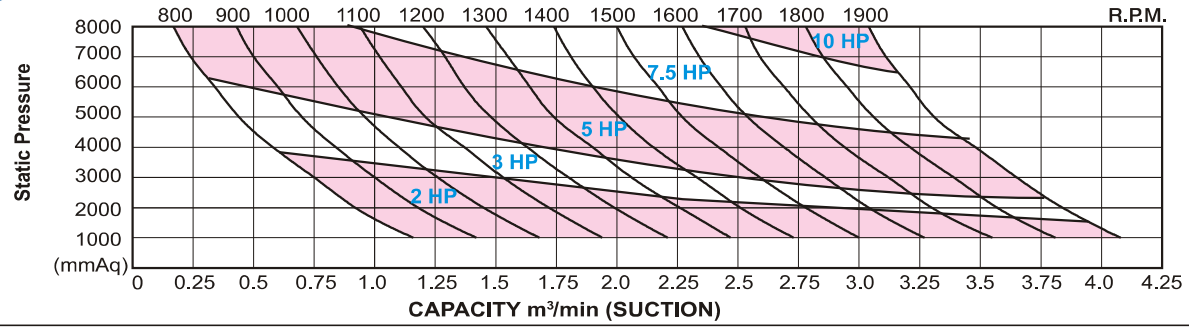
RL PERFORMANCE CURVE (FOR PRESSURE OPERATION)

■ RLC - Complete Set ■ RL - Blower Body only

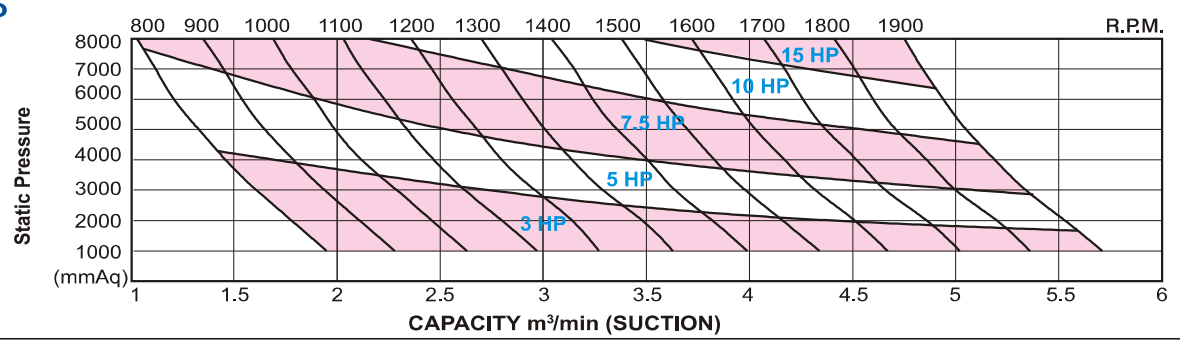
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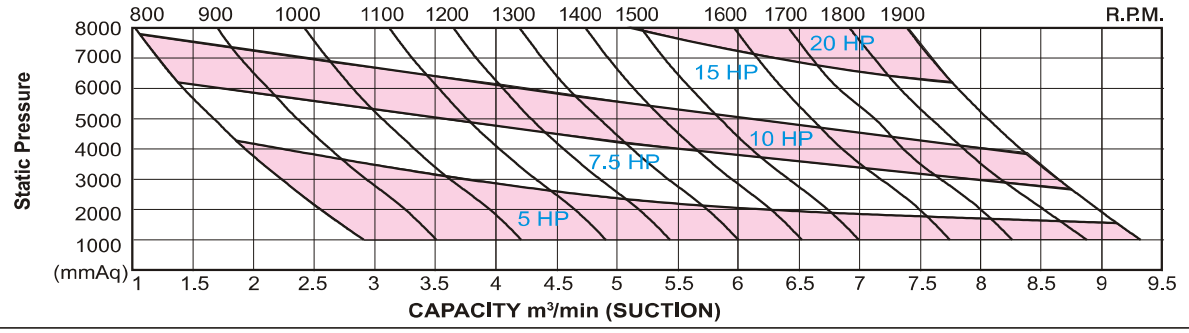
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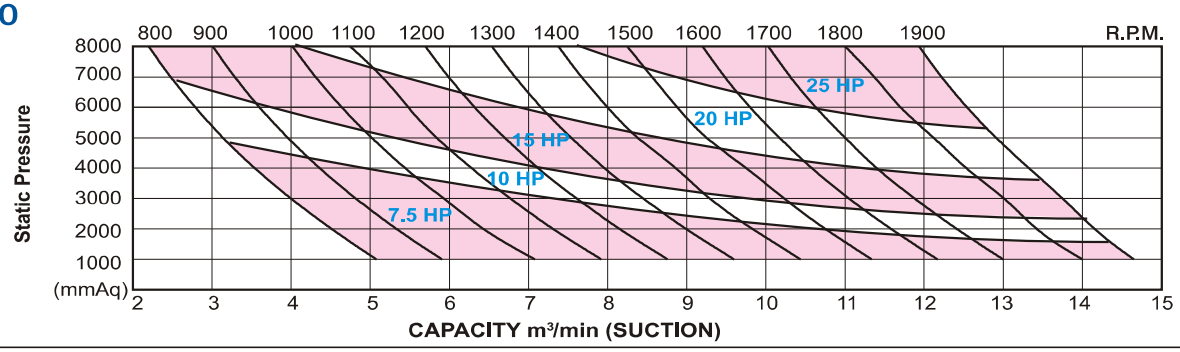
65



80



100

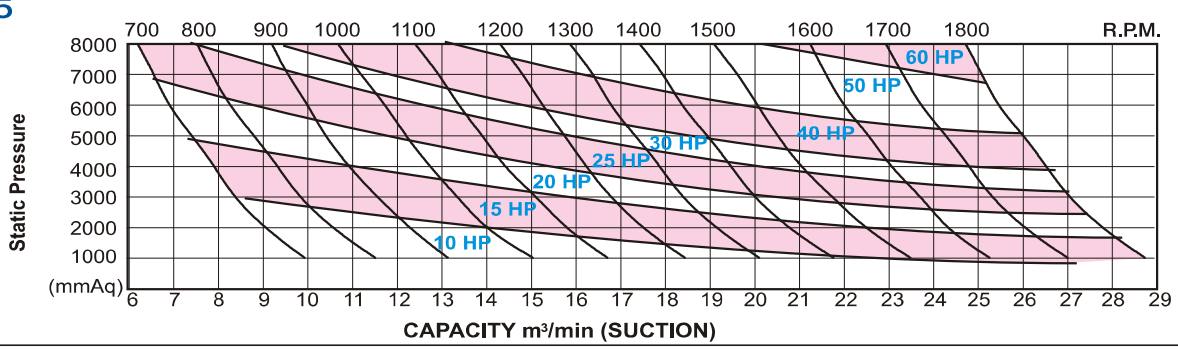


RL

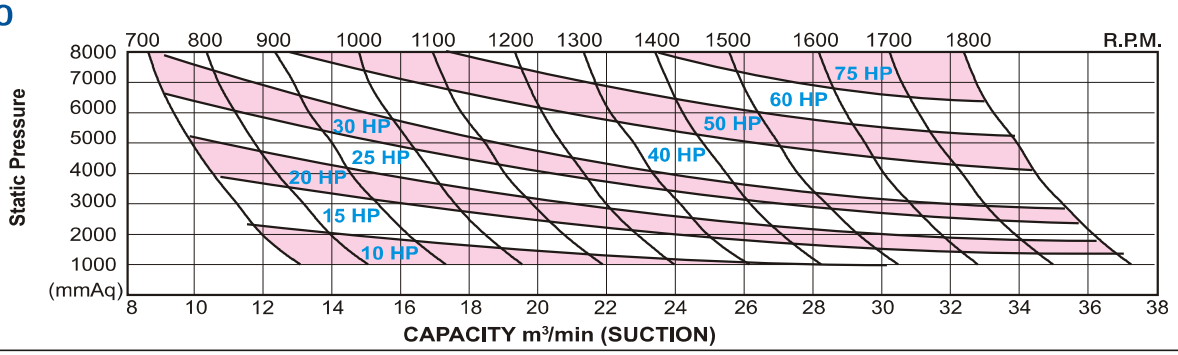
RL PERFORMANCE CURVE (FOR PRESSURE OPERATION)

■ RLC - Complete Set ■ RL - Blower Body only

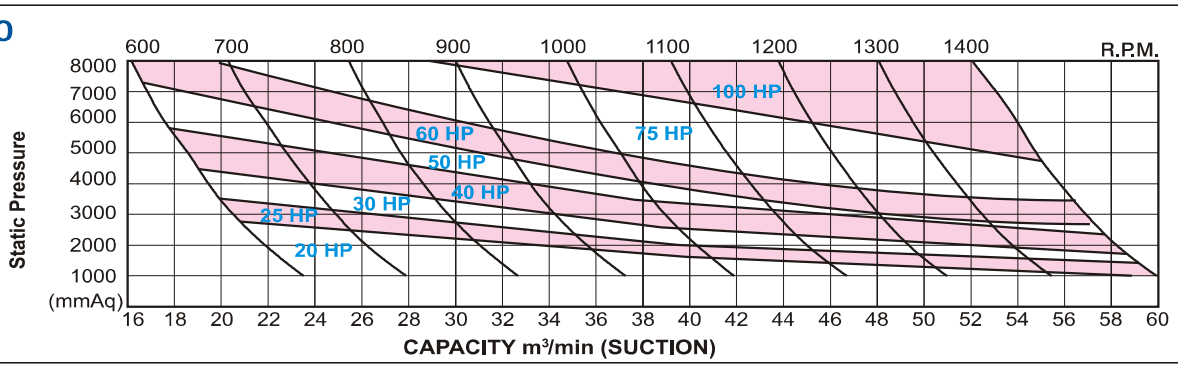
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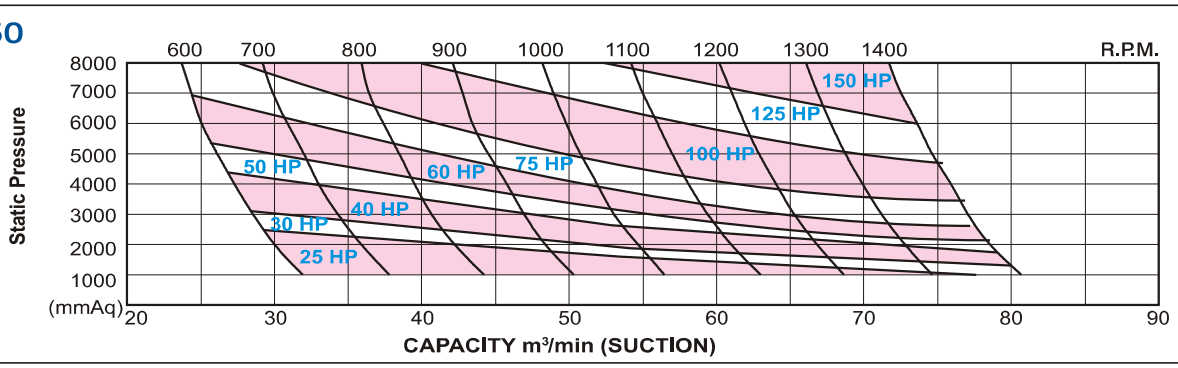
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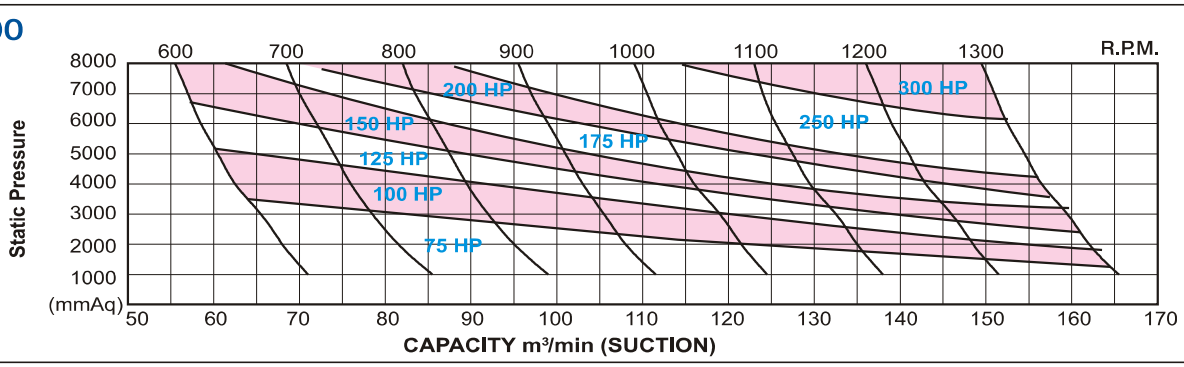
200



250



300



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HOW TO USE PERFORMANCE TABLES

The performance tables give the model number, bore, r.p.m. , static pressure, air volume and required power of the motor.

1. The air volume in the performance table are the volume of air intake under standard conditions.
   Standard conditions are temperature 20 °C, absolute pressure 1 atm (10333mmAq) and relative humidity 65%.
2. Air volume is generally shown in terms of following three modes. Values can be converted between them.
   1) Air volume at **suction condition**
   Shown in terms of suction pressure, temperature and humidity.
   2) Air volume at **standard condition**
   3) Air volume at **base condition**
   When the temperature is 0 °C, absolute pressure 1 atm and when the air is dry.
3. The air volume conversion formula is as follows :
   $Q_2 = Q_1 \times P_1 / P_2 \times T_2 / T_1$
   Q_1 : air volume(m^3/min)at absolute pressure of P_1 (mmAq)
   and absolute temperature of T_1 (K).
   Q_2 : air volume(m^3/min)at absolute pressure of P_2 (mmAq)
   and absolute temperature of T_2 (K).
4. According to the air volume and static pressure as calculated above, the moldel number, bore, r.p.m. and required power can be found in the performance table.
5. The choice is overlapped depending upon the type of blower. For reference, however, selection should be lower number blowers for the economy and higher number for the sound level.
6. The values for power displayed in the performance table are expressed as required power including drive power for the transmission unit, so there is no need to create a margin when selecting a motor.
7. power (output) can be calculated using the following formula
   $kW = \sqrt{3} \times V \times A \times \cos\theta \times Eff. \times 10^{-3}$
   V : voltage A : current cos. θ : motor power factor
   Eff. : motor efficiency

PRESSURE	atm	kpa	mbar	ℓbf/m²(psi)	kgf/cm²	in Hg	ft Aq	mmHg(Torr)	mmAq
1 atm	1	101.325	1013.25	14.696	1.0333	29.921	33.914	760	10333
1 kpa	0.0099	1	10	0.145	0.0102	0.295	0.335	7.5	102
1 mbar	0.00099	0.1	1	0.0145	0.00102	0.0295	0.0335	0.75	10.198
1 ℓbf/m²(psi)	0.069	6.894	68.965	1	0.0703	2.036	2.308	51.71	703
1 kgf/cm²	0.968	98.062	980.392	14.228	1	28.96	32.82	735.53	10000
1 inHg	0.0334	3.3863	33.898	0.491	0.0345	1	1.133	25.4	345.3
1 ftAq	0.0295	2.99	29.851	0.434	0.0305	0.882	1	22.42	304.8
1 mmHg(Torr)	0.013	0.1338	1.333	0.019	0.0014	0.04	0.045	1	13.6
1 mmAq	0.000097	0.0098	0.09803	0.0014	0.0001	0.003	0.003	0.074	1

CAPACITY	m³/min	ℓ/min	cm³/sec	in³/sec	ft³/min(CFM)	U.S. GPM
1m³/min	1	1000	16667	1016	35.288	264.172
1ℓ/min	0.001	1	16.67	1.02	0.0353	0.2641
1cm³/sec	0.00006	0.06	1	0.061	0.002	0.0158
1in³/sec	0.00098	0.983	16.39	1	0.035	0.2589
1ft³/min(CFM)	0.028	28.32	471.957	28.8	1	7.4836
1U.S. GPM	0.003785	3.4785	63.0915	3.8460	0.13360	1

Pressure Conversion Formula

1 kpa=1000 pa=1000 N/M²

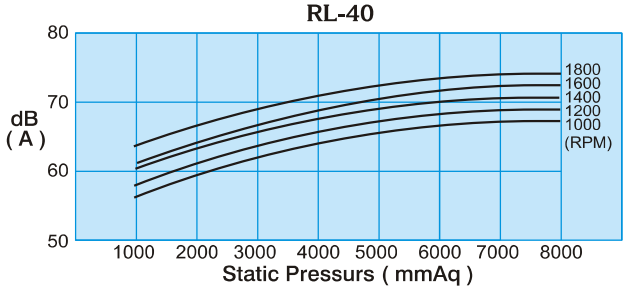
1 bar=1000 mbar=100 kpa

1 mbar=100 N/M²

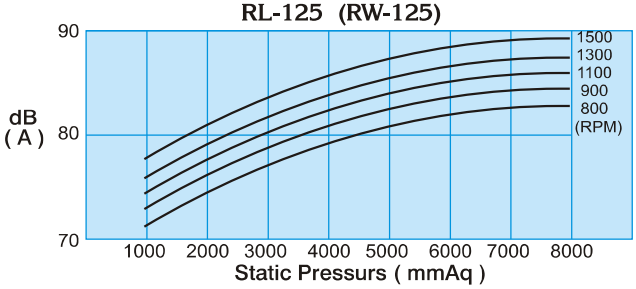
NOISE LEVEL - RL、RW TYPE

1. **Measure method (1.0m On Machine Side)**
   For your information, the following noise reading are the average value measured at a distance 1m from the SHOWFOU blowers under standard condition.
2. **Noise prevention method**
   The noise of SHOWFOU blower has direct relationship with the factors such as blower size, discharge pressure and R.P.M. To get even greater effectiveness in noise reduction, please refer to some methods as follow :
   1. Establish the silencer to inlet and outlet to prevent the noise of air flow.
   2. Decrease the revolution of blower to reduce the noise of it.
   3. Use the expansion joint or the rubber vibration isolator to prevent the surging and vibration.

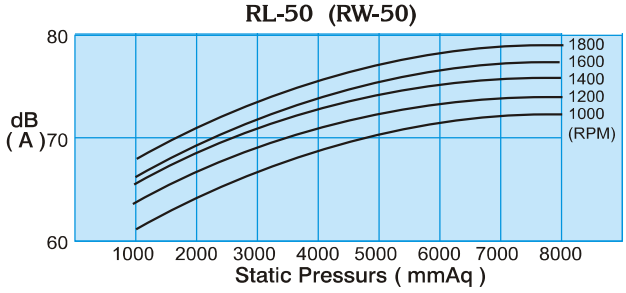
RL-40



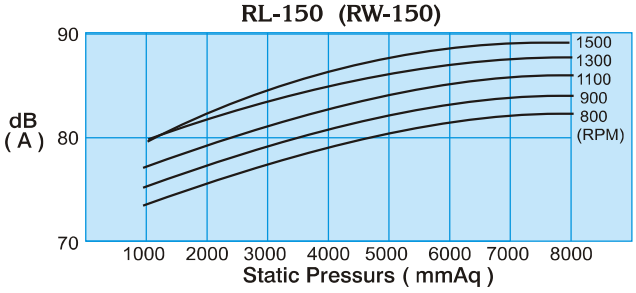
RL-125 (RW-125)



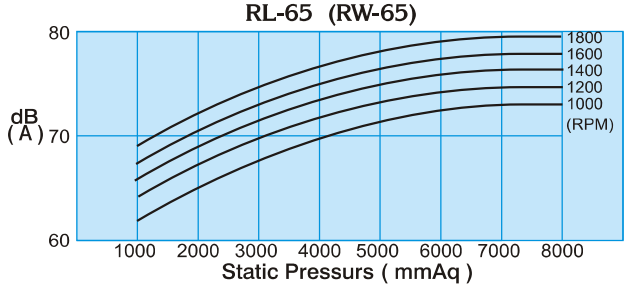
RL-50 (RW-50)



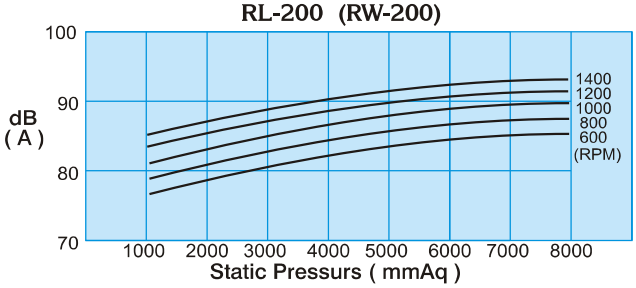
RL-150 (RW-150)



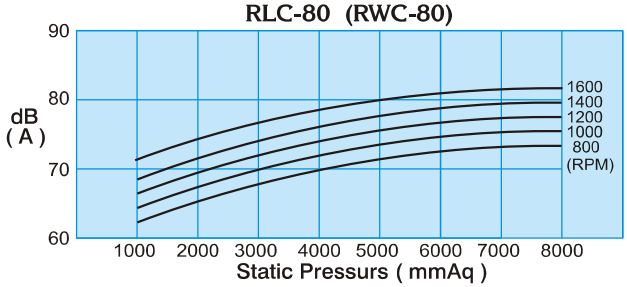
RL-65 (RW-65)



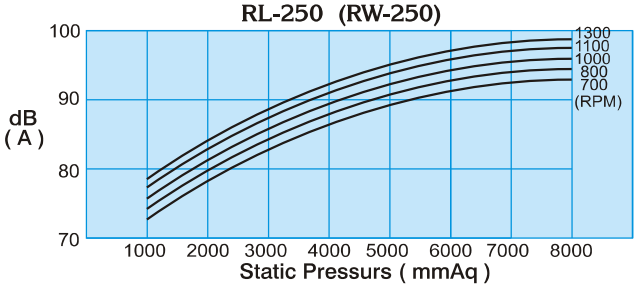
RL-200 (RW-200)



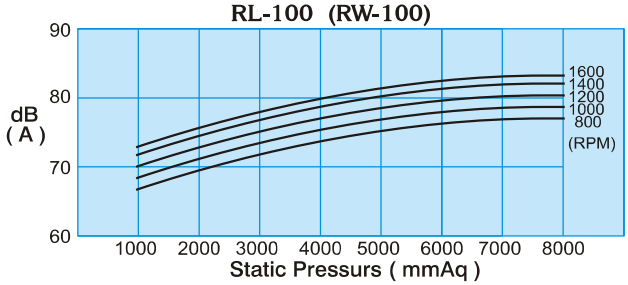
RLC-80 (RWC-80)



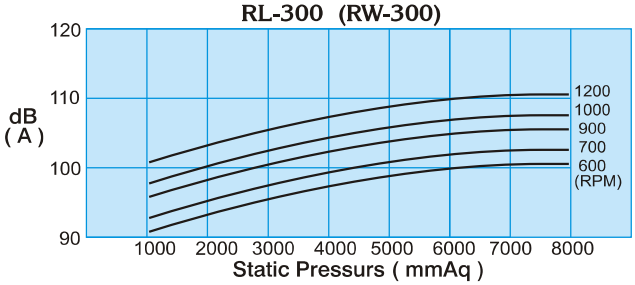
RL-250 (RW-250)



RL-100 (RW-100)

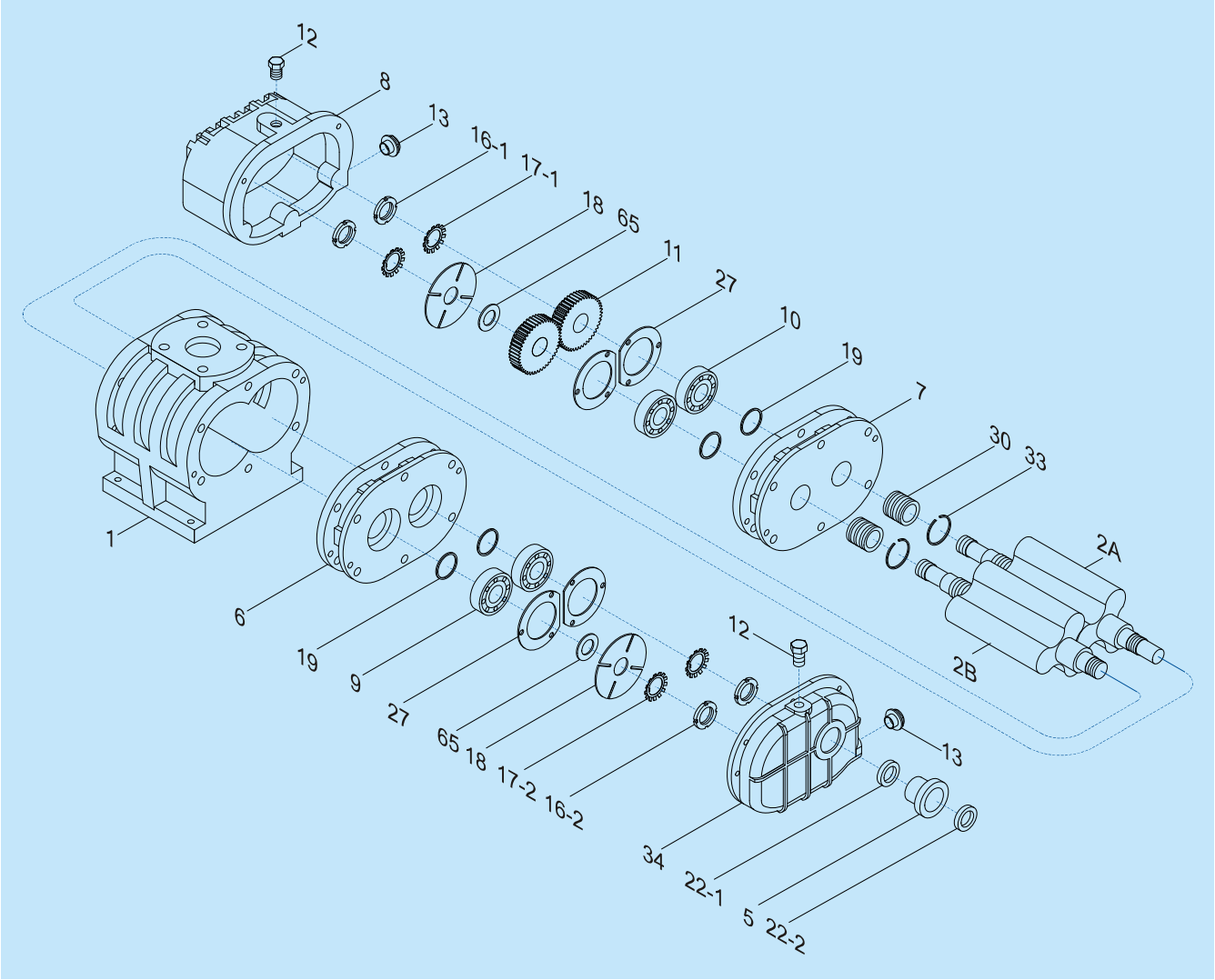


RL-300 (RW-300)



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RL STRUCTURAL DRAWING



PARTS LIST

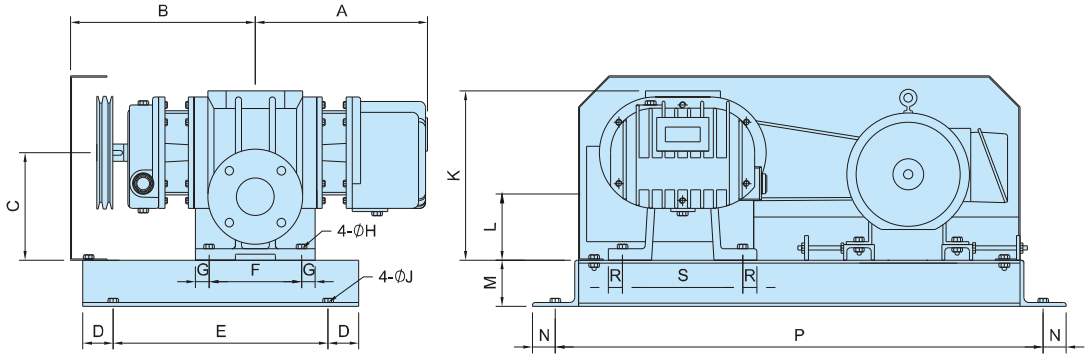
NO.	NAME	MATERIAL	Q'ty	FITTING MODEL
1	CASING	FC25	1	
2A	ROTOR+DRIVE SHAFT	FC25+SCM440	1	
2B	ROTOR+DRIVEN SHAFT	FC25+SCM440	1	
5	SEAL COVER	FC25	4	RL-80~300
6	FRONT SIDE COVER	FC25	1	
7	REAR SIDE COVER	FC25	1	
8	GEAR BOX	FC25	1	
9	FRONT BEARING	SUJ2	2	
10	REAR BEARING	SUJ2	2	
11	TIMING GEAR	SNCM220	2	
12	LUBRICATION PLUG	S45C	2	
13	OIL GAUGE	GLASS	2	

NO.	NAME	MATERIAL	Q'ty	FITTING MODEL
16-1	LOCK NUT	S45C	2	
16-2	LOCK NUT	S45C	2	
17-1	SPRING WASHER	SS41	2	
17-2	SPRING WASHER	SS41	2	
18	OIL SPLASH	SS41	2	
19	V SEAL	VITON	4	
22-1	OIL SEAL	VITON	1	RL- 40~300
22-2	OIL SEAL	VITON	1	RL-125~300
27	BEARING COVER	SS41	4	
30	BEARING HOLDER	SCM440	2	RL-150,RL-300
33	PISTON RING	FC25	2	(OPTIONAL)
34	FRONT COVER	FC25	1	
65	WASHER	SS41	2	

OUTLINE DIMENSIONS

RL TYPE

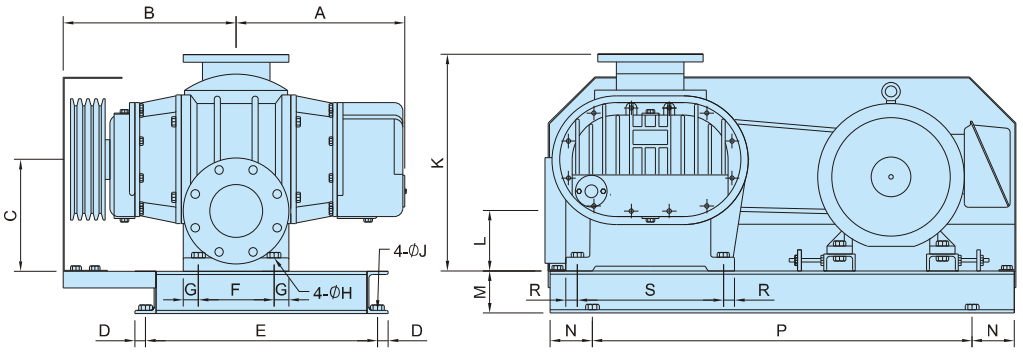
◆ DIMENSIONS OF RL-40,50, 65



UNIT : mm

TYPE	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S	Wt.(kg)	RLC(kg)
RL-40	238	260	180	20	340	80	15	14	14	290	104	46	41.5	517	23	196	65	103
RL-50	278	302	175	50	350	150	22.5	14	14	276	103.5	75	37.5	795	22	196	70	136
RL-65	308	323	173	50	350	195	30.5	14	14	276	104	75	37.5	795	25	200	75	155

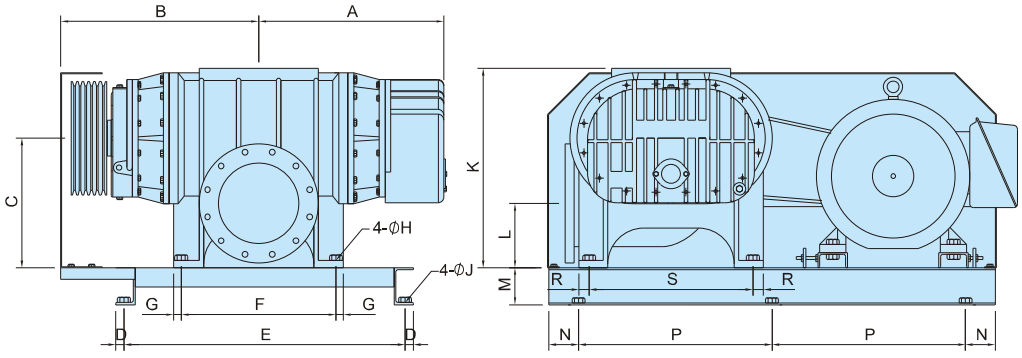
◆ DIMENSIONS OF RL-80, 100, 125, 150



UNIT : mm

TYPE	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S	Wt.(kg)	RLC(kg)
RL-80	320	406	213	50	450	195	22.5	16	14	341	106.5	75	37.5	945	20	200	143	250
RL-100	320	467	225	25	550	282	20.5	16	19	354	114.5	100	100	760	30	232	166	330
RL-125	455	500	265	25	550	310	30	19	19	498	157.5	100	100	900	22.5	355	296	485
RL-150	510	566	295	25	550	428	26	19	19	457	163	100	100	900	22.5	355	352	525

◆ DIMENSIONS OF RL-200, 250, 300



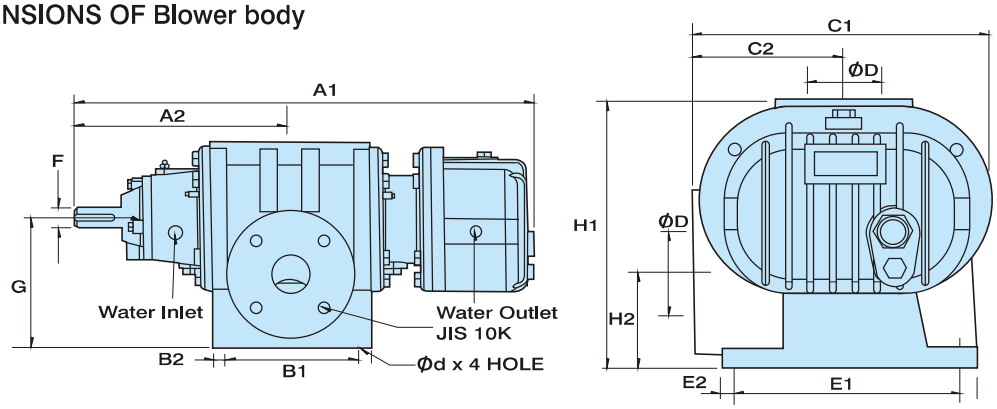
UNIT : mm

TYPE	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S	Wt.(kg)	RLC(kg)
RL-200	547	620	405	32.5	935	400	40	23	24	705	194	125	100	650	33	550	880	1155
RL-250	622	680	435	32.5	935	520	25	23	24	675	216	125	100	650	35	550	1050	1500
RL-300	685	828	550	35	1035	510	30	23	24	1010	300	250	235	1000	35	780	2000	2800

OUTLINE DIMENSIONS

RW TYPE

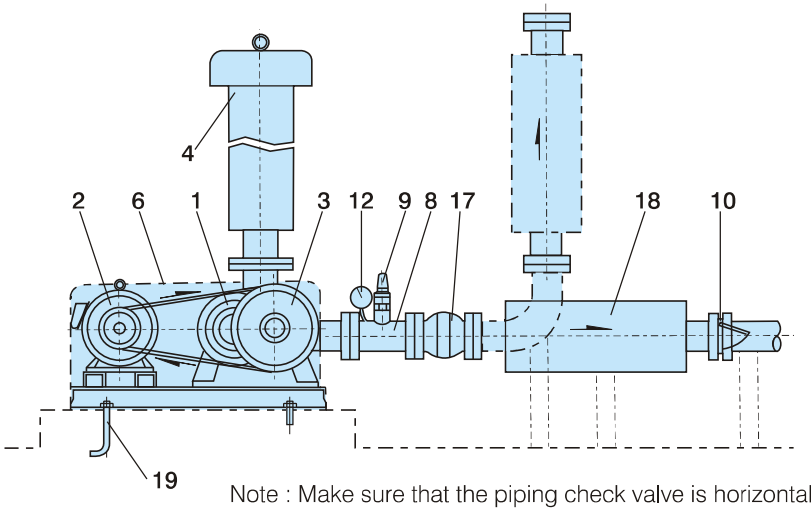
◆ DIMENSIONS OF Blower body



UNIT : mm

TYPE	A1	A2	B1	B2	C1	C2	ϕD	ϕd	E1	E2	F	G	H1	H2	Wt. (kg)	RWC(kg)
RW-50	582	277	150	22.5	294	158	50	14	196	22	28	175	276	104	70	136
RW-65	642	307	195	30.5	288	152	65	14	200	25	28	175	276	104	75	155
RW-80	705	305	195	22.5	358	180	80	16	200	20	38	213	341	107	143	250
RW-100	805	355	282	20.5	364	186	100	16	232	30	38	225	354	115	166	330
RW-125	900	437	310	30	483	250	125	19	355	22.5	48	265	498	158	296	485
RW-150	1010	492	428	26	483	250	150	19	355	22.5	48	295	457	163	352	525
RW-200	1137	590	400	40	685	345	200	23	550	33	65	405	705	194	880	1155
RW-250	1287	665	520	25	715	356	250	23	550	35	65	435	675	216	1050	1500
RW-300	1330	645	510	30	1060	560	300	33	780	35	95	550	1010	300	2000	2800

REFERENCE DRAWING FOR PIPING (EXAMPLE)



Note : Make sure that the piping check valve is horizontal.

NO	ACCESSORIES
1	Blower body
2	Motor
3	Blower pulley
4	Suction silencer set
6	Belt cover
8	T-type reducer
9	Pressure safety valve
10	Check valve
12	Pressure gauge
17	Expansion joint
18	Discharge silencer
19	Anchor bolt

An example of piping arrangement around the blower is shown. Make the most suitable installation from the viewpoint of space, piping schedule, vibration prevention, etc. It is effective to use rubber vibration isolator and to provide an expansion joint to the blower outlet for the prevention of vibration. A silencer may be used to either laterally or vertically.
Install the check valve on the horizontal piping. Use a suitable support to the piping to avoid applying a piping load to the blower nozzle.

DIMENSIONS OF ACCESSORIES

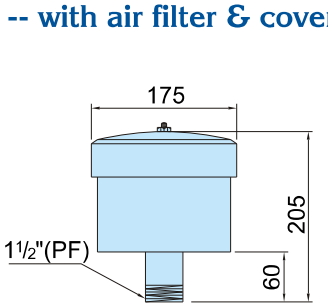
BLOWER PULLEY (No.3)

TYPE	BORE	DIMENSION(O.D.)
SPB170 x 1G	40A	170mm
SPB180 x 2G	50A	180mm
SPB180 x 2G	65A	180mm
SPB200 x 3G	80A	200mm
SPB200 x 4G	100A	200mm

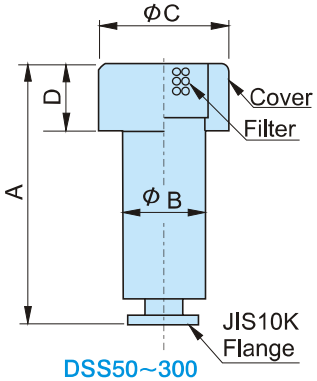
TYPE	BORE	DIMENSION(O.D.)
SPB280 x 4G	125A	280mm
SPB280 x 4G	150A	280mm
SPB375 x 5G	200A	375mm
SPB375 x 6G	250A	375mm
SPC450 x 8G	300A	450mm

SUCTION & DISCHARGE SILENCER SET (No.4)

-- with air filter & cover



DSS40

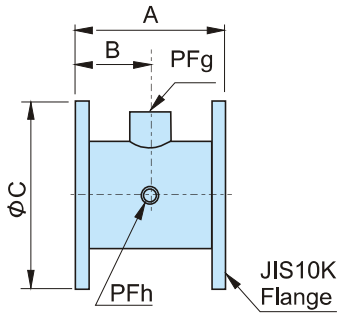


DSS50~300

UNIT : mm

TYPE	BORE	A	B	C	D	WT.(kg)
DSS50	50A	548	140	220	80	7.5
DSS65	65A	548	165	235	85	14
DSS80	80A	818	210	310	145	22
DSS100	100A	818	240	350	160	26
DSS125	125A	1310	280	400	15	30
DSS150	150A	1310	300	450	190	43
DSS200	200A	1500	400	500	210	85
DSS250	250A	1700	400	500	210	110
DSS300	300A	1824	500	650	350	130

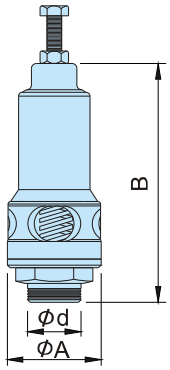
T-TYPE REDUCER (No.8)



UNIT : mm

TYPE	BORE	A	B	C	PFg	PFh	WT.(kg)
TTR50	40A.50A	150	75	155	PF 1 1/2"	PF 1/4"	5
TTR65	65A	150	75	175	PF 1 1/4"	PF 1/4"	7
TTR80	80A	180	90	185	PF 1 1/2"	PF 1/4"	9
TTR100	100A	180	90	210	PF 1 1/2"	PF 1/4"	11
TTR125	125A	250	125	250	PF 1 1/2"	PF 1/4"	18
TTR150	150A	250	125	280	PF 1 1/2"	PF 1/4"	19
TTR200	200A	250	125	330	PF 3"	PF 1/4"	26
TTR250	250A	250	125	400	PF 3"	PF 1/4"	31
TTR300	300A	300	150	450	PF 3"	PF 1/4"	36

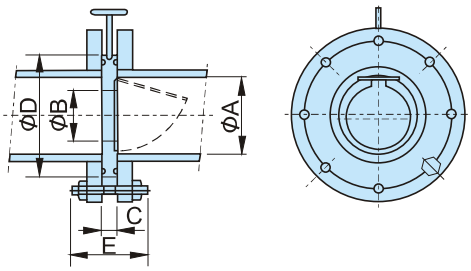
PRESSURE SAFETY VALVE (No.9)



UNIT : mm

TYPE	ΦA	B	Φd	Fitting Model	WT.(kg)
PSV 32	71.5	182	PF1 1/4"	RLC/RWC -40,50,65,80	2
PSV 50	85	197	PF 2"	RLC/RWC -100,125,150	3
PSV 80	115	221	PF 3"	RLC/RWC -200,250,300	5.5

CHECK VALVE (No.10)

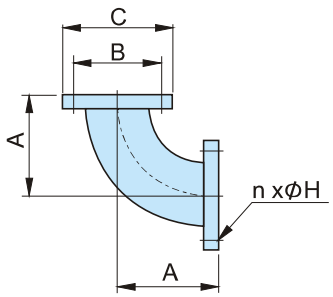


UNIT : mm

TYPE	BORE	A	B	C	D	E	WT.(kg)
NRV50	40A.50A	52.7	25	18	104	86	1
NRV65	65A	69.9	36	19	124	108	1.5
NRV80	80A	78.1	46	20	134	108	2
NRV100	100A	102.3	67	22	159	108	2.5
NRV125	125A	126.6	88	25	190	115	3.5
NRV150	150A	151.0	108	25	220	115	5
NRV200	200A	200.0	138	25	270	127	7
NRV250	250A	251.4	185	26	331	127	10
NRV300	300A	300.0	220	30	370	150	15

DIMENSIONS OF ACCESSORIES

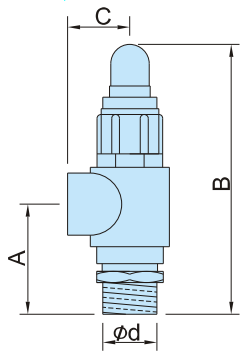
90° ELBOW (No.14)



UNIT : mm

TYPE	BORE	A	B	C	WT.(kg)
NDE50	40A.50A	95.3	120	155	4-Φ19
NDE65	65A	95.3	140	175	4-Φ19
NDE80	80A	114.3	150	185	8-Φ19
NDE00	100A	152.4	175	210	8-Φ19
NDE125	125A	190.5	210	250	8-Φ19
NDE150	150A	228.6	240	280	8-Φ19
NDE200	200A	304.8	290	330	12-Φ19
NDE250	250A	381.0	355	400	12-Φ19
NDE300	300A	457.2	400	445	12-Φ19

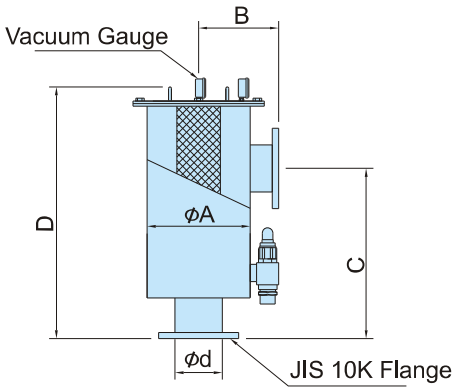
VACUUM SAFETY VALVE (No.15)



UNIT : mm

TYPE	A	B	C	Φd	E	Wt (kg)	FITTING MODEL
VSV 32	83	208	51	32	PT 1 1/4"	1.87	RLV,RWV -40,50,65,80
VSV 50	103	266	67	50	PT 2"	3.91	RLV,RWV-100,125,150,200,250,300

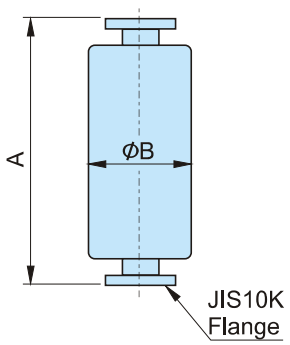
SUCTION FILTER TANK (No.16)
-- vacuum gauge included



UNIT : mm

TYPE	BORE	A	B	C	D	Φd
SFT50	50A	244	222	340	530	60
SFT65	65A	244	222	440	630	76
SFT80	80A	244	222	525	730	89
SFT100	100A	285	240	525	730	114
SFT125	125A	360	280	505	730	139
SFT150	150A	360	280	595	880	165
SFT200	200A	470	335	650	1000	216
SFT250	250A	470	335	800	1200	267
SFT300	300A	600	405	1000	1400	319

DISCHARGE SILENCER (No.18)



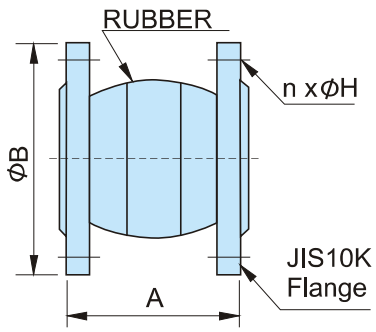
UNIT : mm

TYPE	BORE	A	B	WT.(kg)
DSF50	40A.50A	560	140	12
DSF65	65A	640	165	16
DSF80	80A	850	210	20
DSF100	100A	950	240	28
DSF125	125A	1150	280	33
DSF150	150A	1350	300	41
DSF200	200A	1600	400	90
DSF250	250A	1700	400	120
DSF300	300A	1800	500	140

DIMENSIONS OF ACCESSORIES

EXPANSION JOINT (No.17)

EJA TYPE - RUBBER

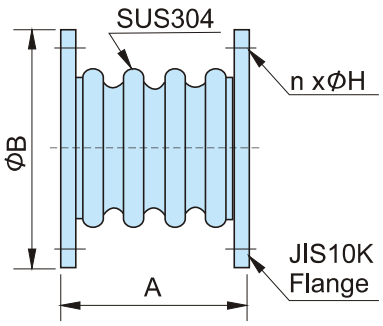


UNIT : mm

TYPE	BORE	A	B	n-ΦH	EJA WT.(kg)
EJA50	40A.50A	85	155	4-Φ19	4
EJA65	65A	95	175	4-Φ19	5
EJA80	80A	110	185	8-Φ19	6
EJA100	100A	115	210	8-Φ19	6
EJA125	125A	140	250	8-Φ19	10
EJA150	150A	150	280	8-Φ19	13
EJA200	200A	170	330	12-Φ 19	16
EJA250	250A	210	400	12-Φ19	18
EJA300	300A	240	445	16-Φ19	21

EJB TYPE - SUS 304

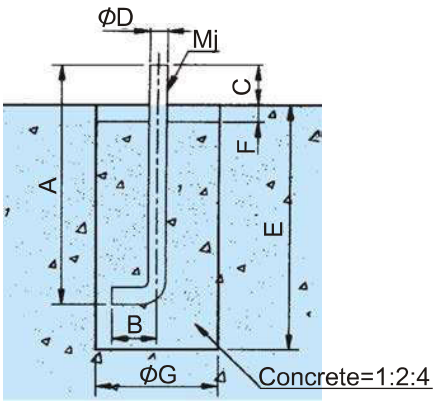
THE EJB TYPE IS RECOMMENDED FOR PRESSURE OVER 5000 mmAq



UNIT : mm

TYPE	BORE	A	B	n-ΦH	EJB WT.(kg)
EJB50	40A.50A	85	155	4-Φ19	4
EJB65	65A	95	175	4-Φ19	5
EJB80	80A	110	185	8-Φ19	6
EJB100	100A	115	210	8-Φ19	6
EJB125	125A	140	250	8-Φ19	10
EJB150	150A	150	280	8-Φ19	13
EJB200	200A	170	330	12-Φ 19	16
EJB250	250A	210	400	12-Φ19	18
EJB300	300A	240	445	16-Φ19	21

ANCHOR BOLT (No.19)



UNIT : mm

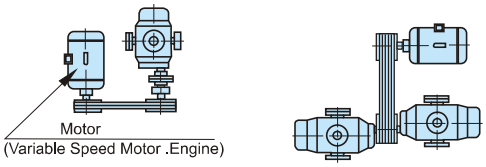
TYPE	Mj	A	B	C	D	E	F	G	FITTING TYPE
AB1	M12	165	30	25	12	165	5	100	RL, RW-50,65,80
AB2	M16	245	45	40	16	245	5	120	RL, RW-100,125
AB3	M20	325	55	55	20	325	5	150	RL, RW-150,200,250,300

perotac
MADE IN TAIWAN

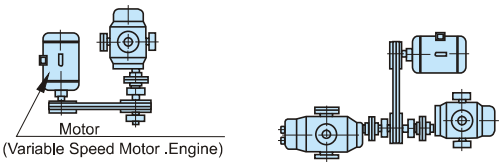
DRIVING METHOD

Motor and V-belt drive is a standard. Blowers with direct coupling to the motor, engine driven blowers with decelerator, or with speed change gear driving may also be adopted. For power in excess of 90 KW(120 HP), the V-belt driving with a counter shaft becomes a standard.

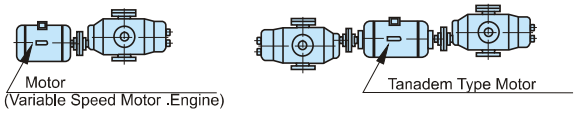
V-belt Driven



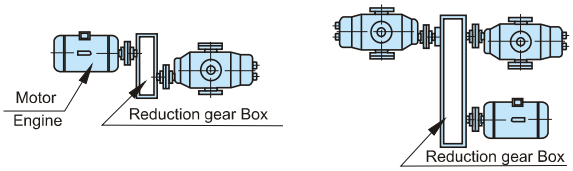
V-belt Counter Shaft Driven



Direct Coupling



Direct Coupling with Reduction Gear Box



APPLICATION GUIDE

The following applications are typical of many uses for SHOWFOU blowers. This is a general listing and does not include all type of applications.

PRESSURE OPERATION

■ Combustion air for burners, gas burners,coke-fired forges, small foundry cupolas in:

- iron & steel industries
- food processing
- chemical plants
- non-ferrous metal refineries
- smelters & foundries
- bakeries
- glass, ceramics, brick
- oil, tar & asphalt heaters
- on road machinery
- pipe, wrapping machines

■ Drying:

- air and sand blasting
- grain conditioning in elevator storage
- supercharging
- dusting and spraying
- washing, drying and cleaning drums
- blending and mixing
- sand filter regeneration
- dust blowing
- cake blowing
- pressurizing paper mill headboxes

■ Aerating and agitating liquids in :

- waste treatment plants
- waste treatment
- electrotyping & engraving
- oyster washing
- oil blending
- ice manufacture
- chemical processes
- preventing ice formation in power plants
- fish ponds, lakes
- flotation
- blending vinegars
- asphalt refining
- food washing

■ Pneumatic conveying:

- granular materials
- insulation blowing
- liquid transfer
- fluidizing
- pulp conveying
- wood and metal chip conveying
- rock dusting
- flour and powder

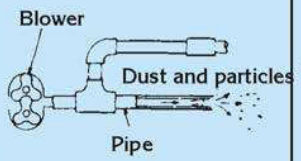
VACUUM OPERATION

- industrial and commercial vacuum cleaning
- paper folding machinery
- garment presses
- bottle and tube filling machines
- drying
- mat formers
- pump priming
- dust collecting

APPLICATIONS

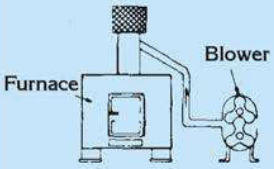
FOR PRESSURE OPERATION

PIPE CLEANING



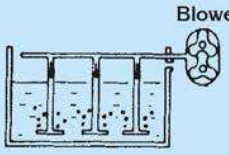
Used to remove dust and rust when replacing piping or conducting periodic inspections.

FURNACE



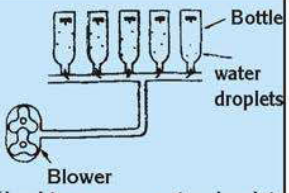
The blower can help boost burn performance and promote exhaust gas removal.

PURIFICATION TANK



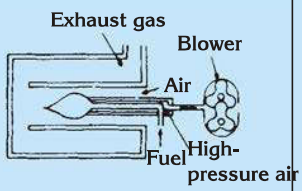
Used to agitate sediment and for purification at water treatment plants.

BOTTLE WASHER



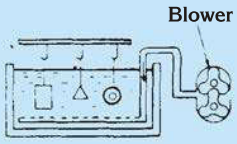
Used to remove water droplets remaining inside bottles after washing.

GAS BURNER



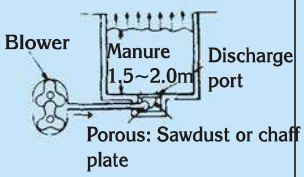
High-pressure discharge air can be used to disperse the fuel.

PLATING TANK



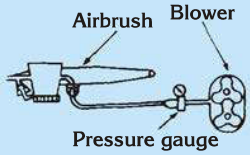
To improve the quality of plating, air is fed into the tank to recirculate the electrolyte. The Roots blower is used as the air source.

FERTILIZER PRODUCTION USING MANURE FERMENTATION



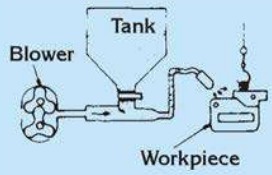
Used to hasten fermentation by feeding air.

AIRBRUSH



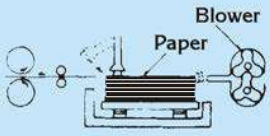
High-pressure discharge air containing no oil is extremely well-received for airbrush applications.

SANDBLASTING



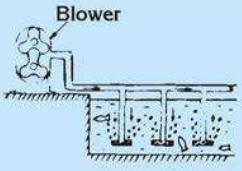
Used as the air source for sandblasting.

PRINTING PRESS PAPER FEED



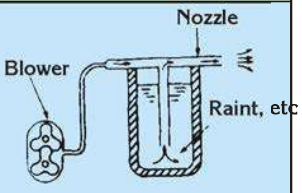
Operations are simplified by using discharge air to separate, align, and feed the paper.

FISHERY POND ENZYME SUPPLY



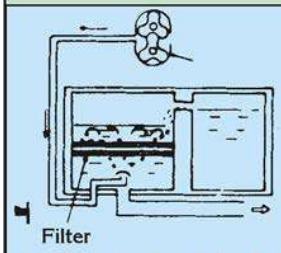
Used for underwater enzyme supply in comparatively shallow ponds.

SPRAYING



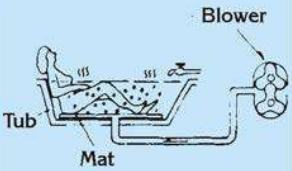
Used to supply high-pressure discharge air containing no oil to the sprayer.

BACKWASH



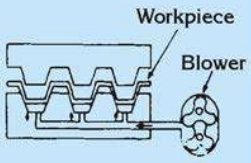
Used as a backwash for filters.

THERAPEUTIC BATH



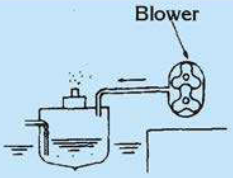
Widely used in hospitals and hotels for healthy bath aeration.

STAMPING DIE



Used to remove the stamped workpiece from the die.

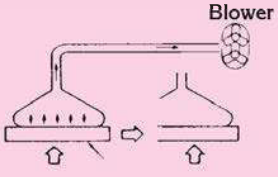
BALLAST PUMP



Used for liguid chemical ship lowering and other types of ballast pumping.

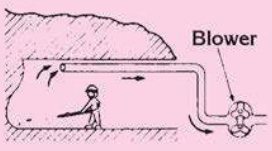
FOR VACUUM OPERATION

ABSORBING CONVEYANCE



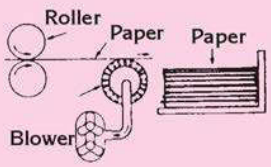
It can convey relatively light materials by absorbing them.

CONSTRUCTION SITE



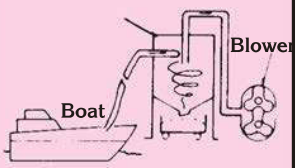
Can be used partial disposal of powders or poisonous gases.

CONVEYING PAPER



Conveying paper in printing machines.

PARTICLES TRANSPORT



Used for air transfer of pellet material.